She Builds Circuits That Talk to Satellites--and She Started by Fixing Toasters!

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Have you ever wondered how your phone talks to a satellite in space? It's all thanks to invisible **radio waves**, and Ana Inês Inácio is one of the engineers making those connections faster and better. She designs super-small **circuits**think smaller than your fingernailthat go into **radar** and future **wireless networks**. Ana didn't start in a fancy lab. She grew up in a tiny village in Portugal where her grandpa fixed looms in a **textile factory**. He taught himself electricity from mail-order books and showed Ana how to repair broken things at home. "He would show me why something broke and how we could fix it," she says. That spark led her to study electronics. Today, she works in the Netherlands at a research center, building the next generation of radio-frequency (RF) sensor systems. Her work helps satellites and sensors "talk" better. She recently won the **IEEE-Eta Kappa Nu** Outstanding Young Professional Awarda huge honor for an engineer under 35. The award praised her for leading teams of young engineers, encouraging new ideas, and making the field welcoming for everyone. Ana says she still loves building things, whether it's **circuits** or communities. Her story shows that curiosity and a knack for fixing things can take you from a village workshop to the edge of space technology.

Word Treasure

radio waves -- Invisible energy waves that travel through the air carrying signals, like from a phone to a satellite.

circuits -- Tiny paths inside devices that let electricity flow to make them work, like a tiny electronic road map.

radar -- A system that uses radio waves to detect objects far away, such as planes or storms.

wireless networks -- Connections that let phones, computers, and other gadgets talk to each other without cables.

textile factory -- A big building where people use machines to turn threads into cloth.

IEEE-Eta Kappa Nu -- A special award given by a huge group of engineers to honor young professionals for outstanding work.

Background you need

Satellites orbit Earth and power GPS maps, TV signals, and weather tracking.

Radio waves, first used for communication by Guglielmo Marconi in the 1890s, carry signals to and from satellites.

The IEEE (the Institute of Electrical and Electronics Engineers, the world's largest professional group for engineers, founded in 1884) gives awards to top young engineers like Ana.

Break it down

WHO: Ana Inês Inácio, an engineer who designs tiny radio-frequency circuits.

WHAT: She builds sensor systems that help satellites and sensors communicate better.

WHERE: She works at a research center in the Netherlands; she grew up in a small village in Portugal.

WHY: Her grandpa taught her to fix broken things, sparking a curiosity that led her to build technology connecting Earth and space.

Why it matters

The invisible radio waves and tiny circuits she designs make things like video calls to friends and GPS navigation possible every day.

Test what you remember

1. **What does Ana Inês Inácio design?**
   - (A) Rockets
   - (B) Super-small circuits
   - (C) Satellites
   - (D) Toasters
2. **Where did Ana grow up?**
   - (A) A big city in the Netherlands
   - (B) A tiny village in Portugal
   - (C) A small town in Spain
   - (D) A village in Brazil
3. **What did her grandpa fix in the factory?**
   - (A) Cars
   - (B) Looms
   - (C) Telephones
   - (D) Computers
4. **In which country does she work now?**
   - (A) Portugal
   - (B) Spain
   - (C) Netherlands
   - (D) United States
5. **What award did she recently win?**
   - (A) Nobel Prize
   - (B) IEEE-Eta Kappa Nu Outstanding Young Professional Award
   - (C) Grammy Award
   - (D) Olympic Medal
6. **What does her work help satellites and sensors do?**
   - (A) Fly higher
   - (B) "Talk" better
   - (C) Look bigger
   - (D) Melt ice

Answer key (try the quiz first!): 1: B 2: B 3: B 4: C 5: B 6: B

Questions to think about

1. Inspiring: Her story shows that a love for fixing things at home can lead to building technology that connects the world.

2. Critical: Not every curious kid has a grandparent with tools or access to advanced electronics classes, so more programs are needed to help them.

3. Neutral: The award recognizes teamwork and community-building, showing that science is also about working together, not just individual genius.

MY THOUGHTS (at least 20 words):